

Title: A satellite-data-driven distributed conceptual hydrological model with mass-conserving-perceptron for the Upper Meghna Basin.

Problem Statement: The Upper Meghna Basin (UMB) is an important transboundary river between Bangladesh and India with limited gauge stations. Data sharing between countries is often difficult due to many reasons. The 65,000 sq. km. UMB is a complex transboundary river basin shared between Bangladesh and India, including wet mountainous regions and deltaic low flood plains, with around 166 large and 6,300 smaller water bodies called “Haors” and “Beels”, respectively (Rahaman et al., 2021). The Meghalaya, Barak, and Tripura regions of India (67% of UMB) are the major hilly sub-basins where heavy seasonal rainfalls occur, which rapidly flash enormous water volumes through the low-lying Sylhet sub-basin of Bangladesh (33% of UMB) (Masood & Takeuchi, 2016).

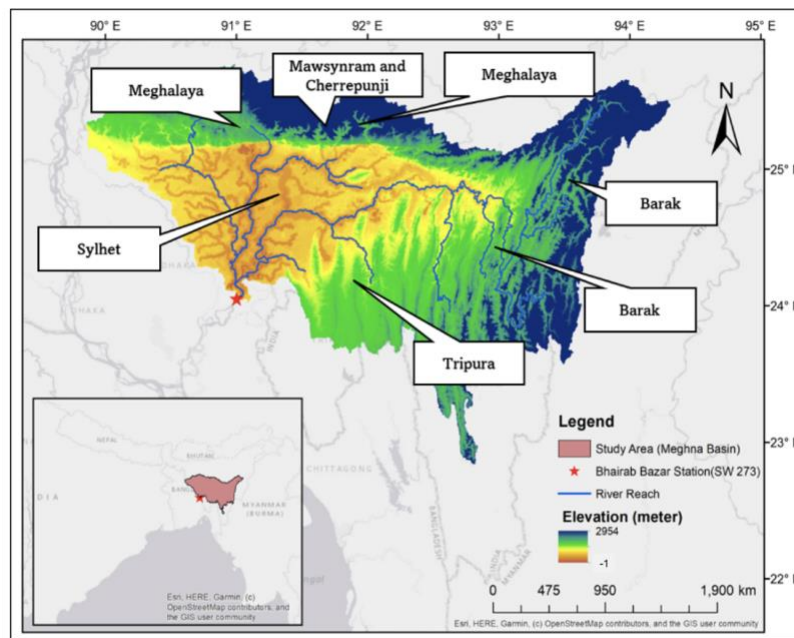


Figure: Topographic map of the Upper Meghna Basin

The basin’s heavy rainfall and geological features give it a flashy hydrological nature, and almost every year the low-lying regions in Bangladesh experience pre-monsoon flash floods and monsoon flooding, causing severe damage to human lives, agriculture, fisheries, and infrastructure (Masood & Takeuchi, 2016). The basin also includes the world’s two wettest regions, Mawsynram and Cherrapunji, which receive an annual rainfall of approximately 11,430 mm, with record daily rainfall reaching up to 1,003 mm and 1,560 mm, respectively (Rahman et al., 2020). Rapid changes in climate are expected to severely alter hydrological behaviors and the flow regimes of the basin (Islam et al., 2018), which may further worsen flood risks in the future. The compound effects of the UMB and two other major basins: the Ganges and Brahmaputra, could lead to devastating impacts in Bangladesh (Gadeke et al., 2022).

Therefore, it is very important to assess climate change impacts on the UMB, particularly for future conditions, to support sustainable policymaking at both national and international levels. One of the key strategies to assess climate change impacts on river basins is to model streamflow, for which hydrological models are commonly used (Islam et al., 2018).

However, recent studies have shown that machine learning and deep learning (ML & DL) models are more efficient and frequently outperform traditional hydrological models in streamflow prediction and forecasting (Lange & Sippel, 2020). Additionally, recent studies indicate that among various DL models, the Long Short-Term Memory (LSTM) model consistently demonstrates superior performance in predicting basin streamflow (Arsenault et al., 2023; Fan et al., 2023; Le et al., 2023; Sabzipour et al., 2023).

Proposed Project: Considering the importance of a highly accurate and fast hydrological model for the UMB, the current project proposes to build a satellite-data-driven MCP based fast distributed hydrological model for that region or least give the fundamental framework upon which future work can be done to refine the model. This will allow engineers, policymakers, water resources managers and others to quickly simulate results, predict near future, analyze climate impact and take appropriate decisions confidently. The main objectives of the current project are:

1. To create mass conserving perceptron or physics based improved perceptron.
2. Use this perceptron and 87 grided precipitation and temperature data to build a conceptual distributed hydrological model for the UMB.
3. Verify the model and publish it for further future work or improvement.

Dataset:

Type	Description	Source/Reference	Resolution and Period
Metrological Dataset	Rainfall Daily	Climate Hazards Group InfraRed Precipitation with Station Data (CHIRPS) (https://www.chc.ucsb.edu/data/chirp)	0.25° 1983-2014
	Max Temperature/Minimum Temperature (Daily)	ERA5 reanalysis Data (https://www.ecmwf.int)	0.25° 1983-2014
Type	Description	Source/Reference	Resolution and Period
Hydrological Dataset	Observed Discharge	Bangladesh Water Development Board	Gauged 1983-2014
	Water Level Data	Bangladesh Water Development Board	Gauged 1983-2014

Methodology:

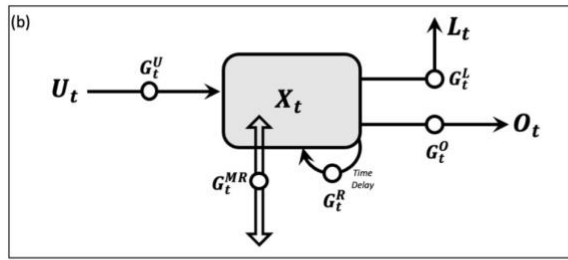


Figure: Directed graph for the generic MCP (Source: Wang & Gupta, 2024)

Output (runoff) flux

$$O_t = G_t^O X_t$$

Loss flux (ET, deep losses)

$$L_t = G_t^L X_t$$

Memory-Return (delayed) flux

$$X_{t+1}^{delay} = G_t^{MR} X_t$$

Full MCP mass balance

$$X_{t+1} = G_t^R X_t + G_t^U U_t + X_t^{delay}$$

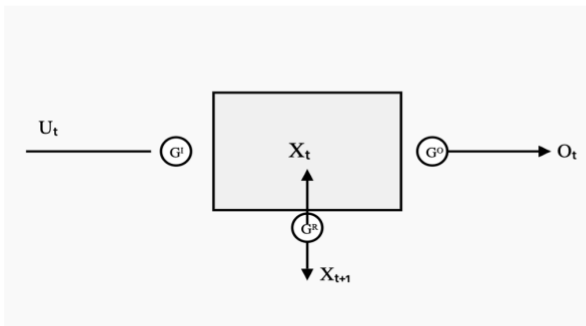


Figure: Directed graph for the MCP used in this study.

Runoff:

$$Q_t = g_O \cdot X_t$$

Updated storage:

$$X_{t+1} = g_R X_t + g_I P_t$$

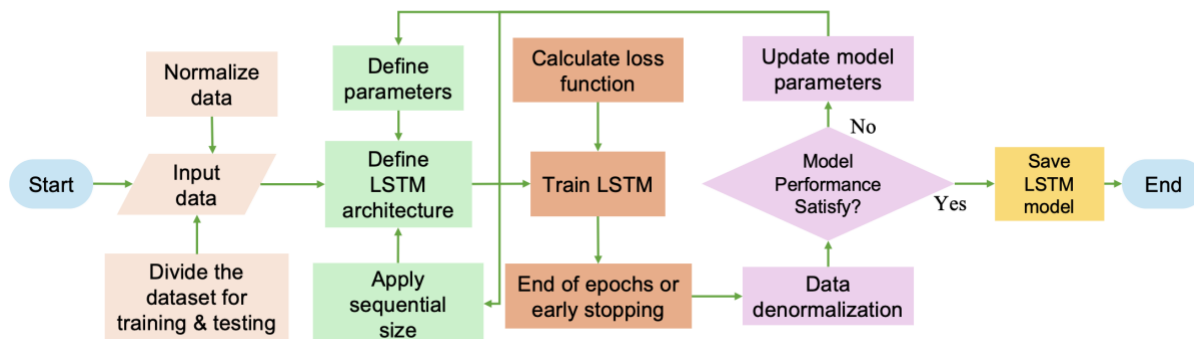
Since all gates are sigmoid:

$$0 \leq g_O, g_R, g_I \leq 1$$

If the network learns:

$$g_O + g_R \leq 1$$

The MCP will be created using gated functions with LSTM perceptron as directed by Wang & Gupta, 2024. Then with that perceptron, first, a lumped model will be created which then will be further refined to a distributed model for each grided/ sub-basin points. Necessary physical concepts will be introduced along the way to better represent the physics in the model.



Expected outcome: An improved representative data-driven fast hydrological model for the UMB that can be used for better decision making and hydrological analysis, rapid simulation and others. This model can be also improved further as future work and can be handed over to the governmental agencies such Bangladesh Water Development Board and the conceptual framework can be published to the scientific community for knowledge sharing.

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